Avidbots™

CPG Validator

LIST OF VALIDATIONS

DOC-00xx Rev 001

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CONTENTS

INTRODUCTION

This document serves as a comprehensive guide for the CPG Validator engine. It details the automated checks, structural constraints, and geometric rules enforced by the system to ensure that generated cleaning plans strictly adhere to the Avidbots Comprehensive Cleaning Plan Generation Guidelines. This application can act as an automated gatekeeper, or in Quality Assurance/ Quality Control phases, since the app informs about malformed SVG files.

Concepts

CPG Tools: Software developed by the engineering team to create and process the SVG file, which is the base file of the cleaning plan used to create all the files that the robot needs to execute a plan. This software is containerized using Docker, and the version of this tool depends on the robot version given by a Docker tag; i.e., the "cpg500_rc0" version is intended to create cleaning plans for the Neo robot, and "kas130_rc0" for the Kas Robot.

Cleaning plan: It consists of multiple files containing instructions for the robots to navigate the facilities.

Once the SVG base file is generated using the CPG tools, the CPG Developers populate it with the information required for the robot to navigate the facilities.

CPG Developers: Are the team members in charge of defining the area that the robot will clean, as well as the order, restricted zones, and different configurations according to guidelines and customer requests. They use the different layers in the SVG file to include this information.

Inkscape: A vector graphics editor software that uses SVG as its main file format. Is used by CPG developers to edit and add information to the SVG file. Please refer to DOC-0046 Rev 001 - Environment and Tools Cleaning Plan Generation Guidelines.

SVG file: Scalable Vector Graphics (SVG). It is a language for describing two-dimensional graphics in XML. SVG supports three types of graphic objects: vector shapes, images, and text. Please refer to DOC-0046 Rev 001 - Environment and Tools Cleaning Plan Generation Guidelines.

XML: Extensible Markup Language is used to parse any SVG file.

Original Map

Validation 1: Original Map Size

The original_map.png size should not exceed 36.5 MPx.

Response error message:

“Please crop the original_map.png image until it gets to less than 35.6 MPx, re-import to plan.svg, and adjust the latter by opening the plan.svg, select the ‘original map layer’, go to File > Document Properties > Display, and enter ‘Resize to conter’.”

Validation 2: Original Map Color

The original_map.png pixels should be colored using one of the following three colors with no transparency:

- White (FFFFFFFF): No obstacles detected by LiDAR
- Black (000000FF): Obstacles detected by LiDAR
- Gray (CDCDCDFF): An area with no LiDAR feedback.

Response error message:

“The original_map.png image should only contain three colors: White (FFFFFFFF), Black (000000FF), or Gray (CDCDCDFF). To fix the issue, you should try:

- Verify your GIMP configurations:
  DOC-0046 Rev 001 - Environment and Tools Cleaning Plan Generation Guidelines
- If there are a few pixels different from the permitted colors, select them and paint them accordingly.
- If there are too many pixels different from the permitted colors, re-copy the original_map.png from the mapping bag and process the image again from scratch.“

Validation 3 - Inkscape Version

The cleaning plan `.svg` file must be generated exclusively using Inkscape version 1.2.2. The validator will scan the XML signature, and files exported from unauthorized software versions or different vector graphic editors will be rejected.

Response error message:

"Open Inkscape and go to Help > About Inkscape, and verify it is in version 1.2.2.

- If yes, open the plan.svg and save the file.
- If no, install the app: https://inkscape.org/release/inkscape-1.2.2/
    - Open the plan.svg and save the file."

Validation 4 - Plan SVG File Size

The total file weight of the SVG should not exceed 10 MB. This prevents network timeouts and high-processing-related issues when transferring the plan to the robotic fleet.

Response error message:

"This SVG file is very heavy, please check:

1. Minimize the original_map.png file size according to the cleanable areas the robot will travel. Remember that the robot only senses up to 20 meters (400 px). You can remove information that the robot will not need in the original map.
2. Make sure you are minimizing the number of logos in the plan image layer.
3. Remove all the NSZs, MOZs, and SDZs that the robot will not use while running the cleaning plan.
4. Remove the real room data that is not relevant to the cleanable area."

Validation 5 - Plan SVG Canvas Dimensions

The canvas area must not exceed 36.5 megapixels (calculated as `width × height <= 36,500,000`). Maps exceeding this limit run the risk of causing high-processing-load crashes on the robot's onboard computers. The validator enforces this with a maximum tolerance of 2 pixels to account for minor Inkscape export artifacts.

Response error message:

"Please crop the original_map.png image until it gets to less than 35.6 MPx, re-import to plan.svg, and adjust the latter by opening the plan.svg, select the 'original map layer', go to File > Document Properties > Display, and enter 'Resize to conter'."

Validation 6 - Plan SVG Canvas Fit

The canvas area must match exactly the original map layer content.

Response error message:

“Please open the plan.svg, select the ‘original map layer’, go to File > Document Properties > Display, and enter ‘Resize to conter’.”

Validation 7 - Layer Encapsulation

All drawn objects (such as rectangles, paths, and text) must be encapsulated within a designated layer.

Response error message:

“The object with ID: <id> is not encapsulated in any layers. Please verify to which layer this object belongs and move it.”

Validation 8 - Plan SVG Global Properties - Part 1

Every SVG file must contain an explicitly defined `environment_type`. The validator will reject the file if this property is missing or if it does not match one of the three approved environments:

- `environment_type: "Standard"`
- `environment_type: "Retail"`
- `environment_type: "Industrial"`

Response error message:

“Please add an environmental type into File > Document Properties > Metadata > Description. Options are: environmental_type: <env_var> | “Standard”, “Retail”, or “Industrial”.

Validation 9 - Plan SVG Global Properties - Part 2

If it is a cleaning plan intended for the Kas robot, the SVG file must contain an explicitly defined `is_homebase` property. The validator will reject the file if this property is missing or if it does not match the following :

- `is_homebase: false`

Response error message:

“This is a Kas cleaning plan, please disable the homebase feature by adding ‘is_homebase: false’ in File > Document Properties > Metadata > Description.”

Validation 10 - Layers’ Existence

The following layers need to exist:

Mandatory layers: ["sectors", "End vector", "Start vector", "distance transform", "cleanable area", "collision check", "annotations", "glass walls", "subplan no go zones", "stealth no go zones", "no go zones", "real room", "original map", "robot image", "move image", "cleaned area", "background"]

Response error message:

“Layer <empty_layer> doesn’t exist, or it’s empty. Please verify.”

Validation 11 - Mandatory Layer Attributes

For a layer group (`<g>`) to be recognized by the system, the validator strictly checks that it explicitly contains the `id`, `inkscape:label`, and `inkscape:groupmode="layer"` attributes.

Response error message:

“Please create the plan.svg from scratch and copy-paste the global metadata and the objects inside layers in the corresponding layers.”

Validation 12 - Layer Transformations Restrictions

The `transform` attribute (such as scaling, matrix rotations, or translations) is strictly prohibited at the layer group (`<g>`) level. Applying transformations to an entire layer visually deceives the map editor while skewing the actual robotic navigation costmap. All coordinates' math and rotations must be applied directly to the individual objects inside the layer, not the layer itself.

Response error message:

“Please remove any transform property in layer: <layer_name>, and adjust the elements inside it.”

Validation 13 - Sectors Layer

The sectors layer defines the areas the robot should clean (Hallways and CHS sectors) or the specific paths it must navigate (Teach and Repeat sectors).

Layer Content: The layer must not be empty. It requires at least one valid object.

Colors: Every object must possess a valid `stroke` color. Allowed colors are Lime (`#00ff00`), Dark Green (`#008000`), Blue (`#0000ff`), and Olive (`#808000`). Unknown colors are rejected.

- **Hallway Sectors**
    - **Trigger:** Stroke color is Lime Green (`#00ff00`), and Fill color = none,
    - **Allowed Shapes:** Only `<rect>` or closed `<path>` tags are permitted.
    - **Transformations:** Allowed.
    - **`<rect>` Rules:** Coordinates (`x`, `y`) must be positive (> 0). Rounded corners (`rx`, `ry`) are strictly prohibited.
    - **`<path>` Rules:** Must be a single stroke (one `M` command), explicitly closed (ends with `Z`), contain no curves, and have exactly 4 nodes. The angles between all pairs of path segments must be exactly 90°.
    - **Minimum Dimensions:**
        - For Neo, a minimum of 5 meters in length and 1 meter in width.
        - For Kas, a minimum of 3 meters in length and 1 meter in width.
        - Do not be square-shaped, meaning: width ≠ height.
- **CHS Sectors**
    - **Trigger:** Stroke color is Dark Green (`#008000`), and Fill color = none.

- **Allowed Shapes:** Strictly limited to straight open <path> tags.
- **Transformations:** Strictly prohibited. The `transform` attribute must be `none`.
- **`<path>` Rules:** Must be a single stroke, explicitly open (no `Z` command), contain no curves, and have at least 2 nodes.
- **Minimum Dimensions:**
  - At least one segment has a length greater than the CHS stroke width.

- **Teach and Repeat Sectors**
  - **Trigger**: Stroke color is Blue (`#0000ff`) or Olive (`#808000`).
  - **Allowed Shapes:** Strictly limited to <path> tags.
  - **Transformations:** Strictly prohibited. The `transform` attribute must be `none`.
  - **`<path>` Rules:** Must be a single stroke, explicitly open (no `Z` command), contain no curves, and have at least 2 nodes.
  - **Dimensions:** maximum 250m (5000px) of length.

Response error message: "There is an error in the 'sectors' layer: <error>."

Validation 14 - End Vector Layer

Specifies the exact position and orientation where the robot will end its autonomous operations.

- **Layer Content:** The layer may contain exactly **1 valid object**. If the layer contains multiple objects, it will be rejected.
- **Allowed Shapes:** The object must be created using an SVG path (`<path>`) and have exactly 2 nodes.
- **`<path>` Rules:** The path must be explicitly open (must not end with the `Z` command) and must not contain any curves (must be a straight line).

Response error message: "There is an error in the 'end vector' layer: <error>."

Validation 15 - Start Vector Layer

Specifies the position where the robot will start autonomous operations.

- **Layer Content:** The layer must contain exactly **1 valid object**. If the layer is empty or contains multiple shapes, it will be rejected.
- **Allowed Shapes:** The object must be created using an SVG path (`<path>`) and have exactly **2 nodes**.
- **`<path>` Rules:** The path must be explicitly open (must not end with the `Z` command) and must not contain any curves (must be a straight line).

Response error message: “There is an error in the ‘Start vector’ layer: <error>.”

Validation 16 - Distance Transform Layer

Post-processed analysis layer generated by the CPG Tools, which tells distances to obstacles in a gray-scale representation.

- **Layer Content:** it must contain exactly **1 valid object**.
- **Allowed Shapes:** The object must be created strictly using an SVG image (`<image>`).
- **`<image>` Rules:** The image must exactly match the `original map` baseline image in the following attributes: `x`, `y`, `width`, and `height`.

Response error message: “There is an error in the ‘distance transform’ layer: <error>.”

Validation 17 - Cleanable Area Layer

Post-processed analysis layer generated by the CPG Tools, which tells the cleanable area of the robot in a white-and-black representation.

- **Layer Content:** it must contain exactly **1 valid object**.
- **Allowed Shapes:** The object must be created strictly using an SVG image (`<image>`).
- **`<image>` Rules:**
  - The image must exactly match the `original map` baseline image in the following attributes: `x`, `y`, `width`, and `height`.
  - The image must contain a white area in all areas on top and around the start point.

Response error message: “There is an error in the ‘cleanable area’ layer: <error>.”

Validation 18 - Collision Check Layer

Post-processed analysis layer generated by the CPG Tools, which tells a number of possible collisions.

- **Layer Content:** it must contain exactly **1 valid object**.
- **Allowed Shapes:** The object must be created strictly using an SVG image (`<image>`).
- **`<image>` Rules:** The image must exactly match the `original map` baseline image in the following attributes: `x`, `y`, `width`, and `height`.

Response error message: “There is an error in the ‘collision check’ layer: <error>.”

Validation 19 - Annotations Layer

This layer is a highly flexible visual container, typically used to map secondary graphical elements, decorative floor plan details, or structured visual assets.

Same validations as ‘no go zones’ layer + <text> validation.

- **<text> Rules:** Text elements must all be set with font-family: sans-serif in their <text> or <tspan> elements.

Response error message: “There is an error in the ‘annotations’ layer: <error>.”

Validation 20 - Glass Walls Layer

This layer is used to highlight transparent glass, reflective materials, or hollow shelves that the robot's LiDAR sensors cannot reliably observe. By annotating these areas, the robot is prevented from attempting to navigate through them, and most important to include these data for its localization algorithm.

- **Allowed Shapes:** Objects must be created using SVG rectangles (<rect>) or paths (<path>).
- **<rect> Rules:**
  - **Minimum Dimensions:** Both the width and height must be ≥ 2 px to prevent microscopic visual artifacts.
  - **Geometry:** The rectangle must have sharp, 90-degree corners. Rounded corners (rx and ry attributes) are strictly prohibited.
- **<path> Rules:**
  - **Straight Lines Only:** The path must not contain any curves (Bézier curves or arcs are forbidden).
  - **Minimum Nodes:** The path must contain at least 2 coordinate nodes.
  - **Closure:** The path may be left explicitly open (for single-plane walls) or closed (for blocks/columns).
  - **Minimum Length:** The maximum distance covered by the path must be ≥ 2 px.

Response error message: “There is an error in the ‘glass walls’ layer: <error>.”

Validation 21 - Subplan No Go Zones Layer

Depending on the current software version and operational guidelines, this layer is reserved for internal testing, legacy compatibility, or future use.

This layer must remain **strictly empty**. If the validator detects any visual elements (such as paths, rectangles, images, or text) inside the layer, the cleaning plan will be immediately rejected to prevent unexpected robot behavior.

Response error message: “The ‘subplan no go zones’ is not empty. Please check if these objects should be moved to the ‘no go zones’ layer”.

Validation 22 - Stealth No Go Zones Layer

Same validation as the ‘no go zones’ layer. This layer defines virtual hard boundaries where the robot is strictly prohibited from entering during its navigation and cleaning routines. The no-go zones in this layer will be hidden from the customer.

Response error message: “There is an error in the ‘stealth no go zones’ layer: <error>.”

Validation 23 - No Go Zones Layer

This layer defines virtual hard boundaries where the robot is strictly prohibited from entering during its navigation and cleaning routines.

- **Allowed Shapes:** must be created using SVG rectangles (<rect>) or paths (<path>).
- **<rect> Rules:**
    - **Geometry:** Rounded corners (rx and ry attributes) are strictly prohibited.
    - **Minimum Dimensions:** Both the width and height must be ≥ 2 px.
    - **Coordinates:** The x and y coordinates must be strictly positive (>0), unless an explicit transform (such as a rotation matrix) is applied to the object.
- **<path> Rules:**
    - **Straight Lines Only:** The path must not contain any curves.
    - **Flexible Topology:** Paths can be explicitly open, explicitly closed (using the Z command), or consist of multiple strokes (multiple M commands) representing fragmented barriers.
    - **Length:** minimum 2-px length.

Response error message: “There is an error in the ‘no go zones’ layer: <error>.”

Validation 24 - Movable Localization Features Layer

Only for Kas plans.

Same validations as glass walls.

Response error message: “There is an error in the ‘movable localization features’ layer: <error>”

Validation 25 - Cloudpointed Zones Layer

This layer designates specific enclosed areas meant for specialized spatial analysis.

- **Allowed Shapes:** Zones must be created strictly using SVG paths (<path>).
- **<path>) Rules:**
  - **Single Stroke:** Each zone must be drawn as a single, continuous object (containing exactly one M command). Multiple disconnected zones must be drawn as separate objects.
  - **Straight Lines Only:** The path must not contain any curves (Bézier curves or arcs).
  - **Strict Closure:** The path must represent an enclosed area and be explicitly closed (ending with the Z or z command).
  - **Polygon Geometry:** The path must contain a minimum of 3 coordinate nodes to successfully form a 2D shape.
  - **Minimum Dimensions:** The maximum physical distance covered by the path nodes must be ≥ 2 px to prevent processing errors caused by microscopic artifacts.

Response error message: “There is an error in the ‘cloudpointed zones’ layer: <error>”

Validation 26 - Static Features Layer

This layer identifies permanent, unmovable physical structures (like architectural pillars, fixed heavy machinery, or built-in counters) that the robot must permanently avoid.

- **Allowed Shapes:** Strict usage of SVG paths (<path>). Other shapes (<rect>, <circle>, etc.) are forbidden.
- **<path> Rules:**
  - **Geometry:** Must be explicitly closed polygons (ending with Z), composed entirely of straight lines (no curves allowed).
  - **Single Stroke:** Each feature must be a single, continuous object.
  - **Minimums:** Must contain at least 3 nodes and have a structural size of ≥ 2 px.
- **Metadata**:
  - There can be no metadata.
  - Or, remove_data: true, or remove_data: false.

Response error message: “There is an error in the ‘static features’ layer: <error>.”

Validation 27 - Movable Obstacles Layer

This layer identifies temporary or movable physical objects (such as pallets, displays, or cleaning carts) that the robot should treat as obstacles but may not be permanently present.

- **Allowed Shapes:** Objects must be created using SVG rectangles (<rect>) or paths (<path>).
- **<rect> Rules:**
  - **Geometry:** The rectangle must have sharp corners (no rx or ry attributes).
  - **Minimum Size:** Both width and height must be ≥ 2 px.
- **<path> Rules:**
  - **Curves Allowed:** Unlike static boundaries, paths in this layer **may contain curves** (e.g., to represent cylindrical objects like bins).
  - **Single Stroke:** Each obstacle must be drawn as a single, continuous object (containing exactly one M command). Multiple obstacles must not be merged into a single path.
  - **Flexible Topology:** Paths can be either explicitly closed (with a Z) or explicitly open.
  - **Minimum Size:** The maximum span of the path must be ≥ 2 px.

Response error message: "There is an error in the 'movable obstacles' layer: <error>."

Validation 28 - Camera Settings Zones Layer

Depending on the current software version and operational guidelines, certain template layers are reserved for internal testing, legacy compatibility, or future use.

This layer must remain **strictly empty**. If the validator detects any visual elements (such as paths, rectangles, images, or text) inside the layer, the cleaning plan will be immediately rejected to prevent unexpected robot behavior.

Response error message: "The 'camera settings zones' is not empty. Please check if these objects should be moved to another layer.

Validation 29 - No Clean Zones Layer

This layer is deprecated and must remain **strictly empty**. If the validator detects any visual elements (such as paths, rectangles, images, or text) inside the layer, the cleaning plan will be immediately rejected to prevent unexpected robot behavior.

Response error message: "The 'no clean zones' is not empty. Please check if these objects should be moved to another layer.

Validation 30 - No Scrub Zones Layer

This layer designates specific enclosed operational areas where the robot must dynamically modify its cleaning behavior by stopping the scrubbing mechanism to protect delicate floors and reducing its speed for safety.

- **Allowed Shapes:** Zones must be drawn using explicitly closed SVG paths (<path>) or standard rectangles (<rect>).
- **<rect> Rules:**
  - **Geometry:** Must be standard rectangles without rounded corners (rx and ry must be 0 or omitted).
  - **Coordinates:** The x and y coordinates must be strictly positive (>0), unless modified by an explicit transform attribute (such as a matrix rotation).
  - **Size:** Both width and height must be ≥ 2 px.
- **<path> Rules:**
  - **Strict Closure:** Paths must represent closed physical areas and must end with the Z or z command. Open lines are forbidden.
  - **Straight Lines Only:** The polygon must be constructed purely of straight lines (no curves allowed).
  - **Single Stroke:** Each zone must be a standalone object consisting of a single continuous stroke (exactly one M command). Do not merge multiple zones into a single object.
  - **Minimums:** The path must contain at least 4 nodes to successfully form an area, and its structural size must be ≥ 2 px to prevent processing errors from micro-clicks.
- **Metadata:** These zones can have dynamic metadata for any of the speeds.

Response error message: “There is an error in the ‘no scrub zones’ layer: <error>”

Validation 31 - Slow Down Zones Layer

This layer designates specific enclosed operational areas where the robot must dynamically modify its cleaning behavior by reducing its speed for safety.

Same validations as the No Scrub Zones layer.

Response error message: “There is an error in the ‘slow down zones’ layer: <error>.”

Validation 32 - Gazebo Obstacles Layer

This layer is deprecated and must remain **strictly empty**. If the validator detects any visual elements (such as paths, rectangles, images, or text) inside the layer, the cleaning plan will be immediately rejected to prevent unexpected robot behavior.

Response error message: “The ‘Gazebo Obstacles’ layer is not empty. Please check if these objects should be moved to another layer.

Validation 33 - Real Room Layer

This layer maps the physical perimeter or actual wall boundaries of the operational space (often used for LiDAR bounding and mapping reference).

- **Allowed Shapes:** Boundaries must be drawn using strictly SVG paths (<path>). Rectangles (<rect>) and all other geometric tags are forbidden.
- **Visibility (Opacity):** The objects in this layer are meant for system calculations, not UI rendering. Every path **must be completely transparent** (the object's opacity must be explicitly set to 0%).
- **Transformations:** The transform attribute is **strictly prohibited**.
- **<path> Rules:**
    - **Strictly Open:** Paths must be explicitly OPEN. Ending the path with a Z or z command will trigger an immediate rejection.
    - **Straight Lines Only:** The boundaries must be constructed of straight line segments. Curves (Bézier curves, arcs, etc.) are not allowed.
    - **Single Stroke:** Each boundary segment must be a continuous single stroke (containing exactly one M command).
    - **Geometry Minimums:** Must contain at least 2 nodes to successfully form a line and cover a physical distance of ≥ 2 px.

Response error message: “There is an error in the ‘real room’ layer: <error>.”

Validation 34 - Original Map Layer

This layer contains the baseline rasterized image of the facility's floor plan, serving as the foundational reference for all operational zones and the robot's mapping engine.

- **Element Count:** The layer must contain **exactly one (1)** object. If the map consists of multiple images, they must be merged or flattened into a single image file before placing them in Inkscape.

- **Allowed Shapes:** Strictly restricted to embedded SVG images (`<image>`).

Response error message: “There is an error in the ‘original map’ layer: <error>.”

Validation 35 - Robot Image Layer

This layer is used to place the graphical icon representing the robot and to hold the elliptical visual footprint.

Element Cardinality: The layer is strictly limited to a maximum of two objects:

  - Exactly **one (1) maximum `<image>`** element (the robot icon).
  - Exactly **one (1) maximum `<ellipse>`** element (the yellow highlight).

Response error message: “There is an error in the ‘robot image’ layer: <error>.”

Validation 36 - Move Image Crop Box Layer

This layer acts as an absolute bounding box, defining the precise coordinate area that will be cropped or rendered for the "move image" visual assets.

- **Element Count:** If populated, the layer must contain **exactly one (1)** object.
- **Allowed Shapes:** Strictly restricted to a standard SVG rectangle (<rect>).
- **Transformations:** The transform attribute is **strictly prohibited**.
- **Containment (Cross-Layer Verification):** The crop box **must mathematically contain the start vector**. The system will automatically scan the start vector layer and reject the map if the starting coordinate falls outside the physical bounds of this crop box.

Response error message: “There is an error in the ‘move image crop box’ layer: <error>.”

Validation 37 - Move Image Layer

This layer is used to define directional indicators (lines) and their associated textual labels (e.g., displaying move distances).

- **Allowed Shapes:** Objects are strictly restricted to standard SVG text (`<text>`) and open paths (`<path>`). Rectangles (`<rect>`) and all other geometric tags are forbidden.
- **Text (`<text>`) Rules:**
  - **Coordinates:** Must possess explicit `x` and `y` baseline coordinates.
  - **Transformations:** Transformations such as `matrix` or `scale` are permitted for text objects.
- **Path (`<path>`) Rules:**

- **Topology:** The path must be an explicitly **open line** (no `Z` command) and consist of a **single stroke** (one `M` command).
- **Geometry:** Must be constructed purely of straight lines (no curves).
- **Exact Nodes:** The path must contain **exactly two (2) coordinate nodes** (representing a precise start and end point). Paths with 3 or more nodes will be rejected.

Response error message: "There is an error in the 'move image' layer: <error>."

Validation 38 - Cleaned Area Layer

Post-processed analysis layer generated by the CPG Tools, which tells the cleaned area of the robot in a white-and-black representation.

- **Layer Content:** it must contain exactly **1 valid object**. If it contains multiple objects, it will be rejected.
- **Allowed Shapes:** The object must be created strictly using an SVG image (`<image>`).
- **Geometry Rules:** The image must exactly match the `original map` baseline image in the following attributes: `x`, `y`, `width`, and `height`.

Response error message: "There is an error in the 'cleaned area' layer: <error>."

Validation 39 - Plan image Layer

This layer is a highly flexible visual container, typically used to map secondary graphical elements, decorative floor plan details, or structured visual assets.

- **Allowed Shapes:** Broadly supports most standard SVG visual tags: <rect>, <circle>, <ellipse>, <path>, <text>, and <image>.
- **Transforms:** All forms of spatial transformations (matrix, scale, rotation) are fully permitted.
- **Topology:** Paths can be curved, straight, open, closed, single-stroke, or multi-stroke.
- **Corners:** Rectangles are permitted to have rounded corners (rx and ry).

Response error message: "There is an error in the 'plan image' layer: <error>."

Validation 40 - Background Layer

This layer defines a solid background color for any PNG image export.

- **Element Cardinality:** If populated, the layer must contain **exactly one (1)** element.
- **Allowed Shapes:** Strictly restricted to a single standard SVG rectangle (<rect>). Images, paths, and groups are strictly prohibited.

- **Geometry Rules:**
  - **Coordinates:** The rectangle must be perfectly anchored to the canvas origin (both x and y must be exactly 0).
  - **Transformations:** The transform attribute is **strictly prohibited** (transform="none").
  - **Dimensions:** Both width and height must be valid, positive values representing the full span of the canvas.

Response error message: "There is an error in the 'background' layer: <error>."

MAPS

Validation 41 - Map3d PNG Size

- Same size as the original map.

Response error message: “The map3d.png is not the same size as original_map.png.”

Validation 42 - Map3d PNG Cleanliness

- No text or logos inside the cleanable area.

Response error message: “The map3d.png presents obstructive elements that do not correspond to the navigable system. Please verify the position of text and logos in the annotations layer.”

Validation 43 - Map PNG Size

- Same size as the original map.

Response error message: “The map.png is not the same size as original_map.png.”

MAIN FILES

Validation 44 - Config.yaml File

- If it is a Kas plan, then the plan type should be 1 (cleaning).

Response error message: “The plan type is different than 1 (cleaning). Please verify the ‘is_homebase: true’ metadata is included in File > Document Properties > Metadata > Description.”

Validation 45 - Sectors.yaml File

- Global properties should be parsed into all sectors, except for pressure, flow, environmental_type, and is_homebase properties.
  - If it is a Kas plan, wall_distance should always be 0.15m.
  - If it is a Neo plan, wall_distance should be between 0.20m and 0.35m.
- Local properties should be parsed only into the corresponding sectors.

Response error message: “There is an error in the sectors.yaml file: <error>”

Validation 46 - Plan.world File

- The files should not be empty.
- The file should not exceed 100 MB in size.

Response error message: “There is an error in the plan.world file: <error>”

CUSTOMER-FACING IMAGES

Validation 47 - Plan PNG Size

- Same size as the original map.

Response error message: “The plan.png is not the same size as original_map.png.”

Validation 48 - Pretty Map PNG Size

- Same size as the original map.

Response error message: “The pretty_map.png is not the same size as original_map.png.”

Validation 49 - Move PNG

- Can be the same size as the original map or smaller.

Response error message: “There is an error in the move.png file: <error>.”

Cleanable Area Images

Validation 50 -Cleanable Area PNG Size

- Same size as the original map.

Response error message: "The cleanable_area.png is not the same size as original_map.png."

Validation 51 -Cleanable Area PNG Color

- The start point and its near surroundings should not be covered by any zones with a cleaning system OFF, such as MOZ or NSZ.

Response error message: "The cleanable_area.png is totally black. Please remove from the start point and its surroundings any zones with the cleaning system OFF, such as MOZ or NSZ."

Document Revision Control

Doc #	Rev	Date	Author(s)	Revision Details
DOC-0052	001	Apr 14, 2026	Luis Vasquez Jaime Andre... Mateo Garcia	Initial revision.